Question

At room temperature, 0.15 g of the compound $\mathbf{H_3L}$ (structure shown in Figure 1) was dissolved in anhydrous ethanol, followed by the addition of 1 mL of pyridine solution and 0.09 g of nickel acetate. The mixture was stirred at 50°C for 30 min to obtain a solution. Slow evaporation in air yielded block-shaped single crystals $\mathbf{X}$ with a 48% yield. Elemental analysis of $\mathbf{X}$ revealed: C 52.94; H 3.33; N 12.87; O

14.69 (all in mass percent, %).

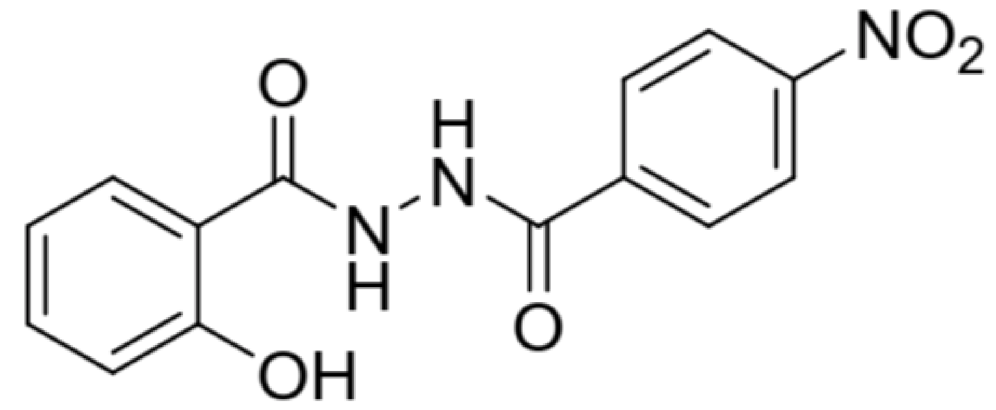

Fig. 1. The SMILES representation of this molecule is: O=C(NNC(C1=CC=C([N+]([O-])=O)C=C1)=O)C2=CC=CC=C2O

The chemical formula of $\mathbf{X}$ was deduced through calculations.

Infrared spectroscopy was performed on both compound $\mathbf{H_3L}$ and the resulting complex $\mathbf{X}$, yielding the following results.

Key IR absorption peaks $\left(\mathrm{KBr, cm^{-1}}\right)$: $\mathbf{H_3L}$: $1692; 1590; 1543; 1520;$ $\mathbf{X}$: $1602; 1563; 1518$.

By analyzing the differences in their IR spectra, the coordination structure of $\mathbf{L}$ was deduced.

Which of the following options is correct?

A. None of the other options are correct

B. In a $\mathbf{X}$ molecule, the product of the total quantities of various atoms is 172800.

C. In a molecule of $\mathbf{X}$, the product of the total number of various atoms is 345600.

D. In a $\mathbf{X}$ molecule, the product of the total numbers of various atoms is 10800.

E. The UV-visible absorption peak of the coordinated $\mathbf{L}$ molecular structure (excluding the absorption of the central Ni) may exhibit a significant red shift.

F. The variation in IR absorption peaks is caused by the formation of coordinate bonds when $\mathbf{L}$ molecules coordinate, leading to the weakening of the original infrared-active four chemical bonds.

G. There is no change in the position of the double bond in the $\mathbf{L}$ molecule before and after coordination.

Answer

Correct Answer: E

Detailed Explanation

From the elemental analysis results, we obtain $n(\mathrm{C}) : n(\mathrm{O}) = 4.80$, $n(\mathrm{C}) : n(\mathrm{H}) = 1.33$, and $n(\mathrm{N}) : n(\mathrm{O}) = 1.00$.

The nitrogen-containing ligands in $\mathbf{X}$ can only be $\mathbf{L}$ and pyridine, with the chemical formulas $\mathrm{C_{14}H_8N_3O_5}$ and $\mathrm{C_5H_5N}$, respectively.

From $n(\mathrm{N}) : n(\mathrm{O}) = 1.00$, it can be deduced that the ratio of $\mathbf{L}$ to pyridine is $1 : 2$.

CHECKPOINT 1 PTS

The ratio of $\mathbf{L}$ to pyridine in the ligands is $1 : 2$

Assuming $\mathbf{X}$ contains 1 $\mathbf{L}$ and 2 pyridines, the number of C atoms would be 24, N and O atoms would each be 5, and H atoms would be 18, which precisely matches the ratios obtained from the elemental analysis. Therefore, $\mathbf{X}$ contains only $\mathbf{L}$ and pyridine as ligands. If the number of $\mathbf{L}$ in $\mathbf{X}$ is $x = 2$, then the number of pyridines is $2x$.

The central metal is clearly Ni. When $x = 2$, the remaining mass $M = 176.1\ \mathrm{g/mol}$ corresponds to 3 Ni atoms. Thus, the chemical formula of $\mathbf{X}$ is $\mathrm{Ni_3}\mathbf{L}_2\mathrm{(C_5H_5N)_4}$, written as $\mathrm{Ni_3C_{48}H_{36}O_{10}N_{10}}$, with a coefficient product of 518400. Options A, B, C, and D are incorrect.

CHECKPOINT 1 PTS

The chemical formula of $\mathbf{X}$ is $Ni_3\mathbf{L}_2(C_5H_5N)_4$, with a coefficient product of 518400

When the compound $H_3\mathbf{L}$ exists alone, it exhibits the equilibrium shown in Figure 2:

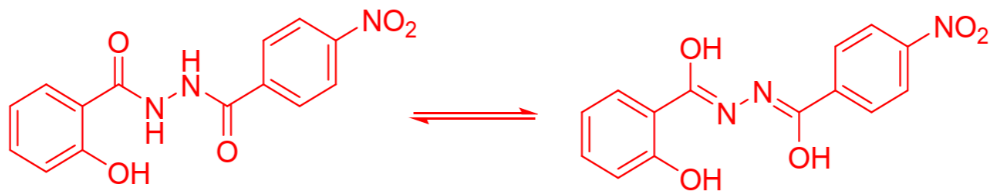

Fig. 2, the figure depicts a reversible reaction: O=C(C1=CC=CC=C1O)NNC(C2=CC=C(C=C2)[N+]([O-])=O)=O>>O=[N+]([O-])C(C=C3)=CC=C3/C(O)=N/N=C(O)/C4=CC=CC=C4O

Thus, $H_3\mathbf{L}$ displays four major infrared absorption peaks. The difference between $\mathbf{X}$ and $H_3\mathbf{L}$ lies in the absence of the carbonyl absorption peak near $1700\ cm^{-1}$, making option F incorrect. When $\mathbf{L}$ coordinates with the central metal, it undergoes keto-enol tautomerization, with the structure shown in Figure 3, making option G incorrect.

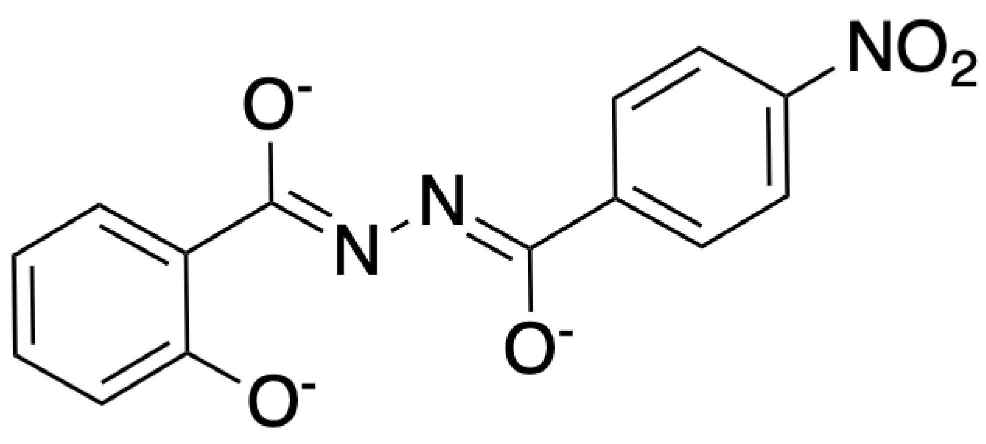

Fig. 3, the molecular structure is represented in SMILES as: [O-]C(C(C=C1)=CC=C1[N+]([O-])=O)=NN=C(C2=CC=CC=C2[O-])[O-]

When coordinated, $\mathbf{L}$ forms a large conjugated system, causing the absorption peak to redshift, making option E correct.

CHECKPOINT 2 PTS

The difference between $\mathbf{X}$ and $H_3\mathbf{L}$ is the absence of the carbonyl absorption peak near $1700\ \mathrm{cm}^{-1}$, and $\mathbf{L}$ forms an enol isomer upon coordination.